

Profile

Theoretical Physicist

Theoretical Physicist with 4+ years of experience in theory of quantum optimal control, specializing in its application to superconducting qubits. Demonstrated expertise in modelling, simulation, and optimization of many-body quantum systems. Collaborated in a team of researchers and software engineers for the development of optimal control package [ParaQeet](#), based on fast linear algebra and automatic differentiation using JAX. Eager to explore methods for efficient simulation and optimization of multi-partite systems involving large Hilbert space dimension.

Education

- **Master of Science in Physics**
Indian Institute of Science
July, 2021
Bangalore, India
 - CGPA: 9.0/10.0
 - First class with distinction
- **Bachelor of Science (Research) in Physics**
Indian Institute of Science
July, 2020
Bangalore, India
 - CGPA: 8.8/10.0
 - First class with distinction

Experience

- **Forschungszentrum Jülich GmbH**
Doctoral researcher
Sep 2021 - Present
Jülich, Germany
 - **Supervisors:** Prof. Dr. Frank Wilhelm-Mauch, Dr. Alessandro Ciani
 - **PhD thesis:** Open-loop quantum control for practical superconducting quantum systems (submitted)

For my PhD, I implemented numerical and analytical open-loop quantum control methods to systems with large Hilbert space and open quantum systems. I demonstrated that:

 - * A general framework for open-loop control can be formalized for arbitrary pulse parameterization.
 - * A series expansion can be derived which, combined with Lieb-Robinson bounds, provides an efficient method for open-loop control in lattice systems.
 - * Using numerical techniques, readout and re-initialization in fixed frequency transmons can be achieved by a *demolition measurement* protocol.

Also, I contribute to the development and maintenance of quantum optimal control package: [ParaQeet](#).
- **Indian Institute of Science**
Bachelor's and Master's thesis
Aug 2016 - July 2021
Bangalore, India
 - **Supervisor:** Dr. Baladitya Suri
 - **Master's thesis:** Estimating errors in two qubit parametric iSWAP gate.
 - * Investigated errors due to frequency crowding in multi-qubit coupler based iSWAP gate implementation.
 - * Performed simulation of 4 qubits and a tunable coupler using Julia.
 - **Bachelor's Thesis:** Input-output theory for qubit measurement in a circuit QED system.

- * Examined input-output relations for one-sided and two-sided cavity model for the readout of a transmon coupled to a resonator.
- * Verified the results by simulation using QuTiP.
- **The Australian National University** May 2019 - July 2019
Research internship Canberra, Australia
 - **Supervisor:** Dr. Ben Buchler
 - Part of visiting scholarship award of *Chennupati and Vidya Jagadish Visiting Scholarship and Fellowship Awards*.
 - * Built a trans-impedance amplifier circuit for a photo-detector used for high speed detection of laser signals.
 - * Studied about quantum memories using electromagnetically induced transparency.

Research projects

- **Analytical Series Expansion for Efficient Gradient Evaluation in Multi-Qubit Optimal Control** Sep 2024 - Present
Supervisors: Prof. Dr. Frank Wilhlem-Mauch, Dr. Alessandro Ciani (PGI-12) Forschungszentrum Jülich
 - Derived an efficient method for gradient computation during open-loop optimization of lattice-based large multi-qubit systems, and open quantum systems.
- **ParaQeet-A quantum optimal control toolkit with simple parameter management** Dec 2024 - Present
Supervisor: Dr. Alessandro Ciani (PGI-12) Forschungszentrum Jülich
 - Contributed to the development of a semi-automatic differentiation based quantum optimal control package, written in JAX.
- **Demolition measurement protocol for transmon qubits** Sep 2021 - Feb 2024
Supervisors: Dr. Shai Machnes (Qruise GmbH), Prof. Dr. Frank Wilhlem-Mauch (PGI-12) Forschungszentrum Jülich
 - Performed numerical investigation into a combined protocol for superconducting qubit readout, leakage removal and unconditional reset.
- **Estimating errors in two qubit parametric iSWAP gate** Aug 2020 - July 2021
Supervisor: Dr. Baladitya Suri (Dept. of Instrumentation and Applied Physics) Indian Institute of Science
 - Investigated error budget due to frequency crowding in a multi-qubit coupler based iSWAP gate implementation using Julia.
- **Study on effect of heterogeneity and plasticity on cell population under effect of drug** May - Aug 2020
Supervisor: Dr. Mohit Kumar Jolly (Centre for BioSystems Science and Engineering) Indian Institute of Science
 - Mathematical modeling and agent based simulation of cell population under drug treatment and studied the effects of heterogeneity and plasticity.
- **Input-output theory for qubit measurement in a circuit QED system** Aug 2019 - May 2020
Supervisor: Dr. Baladitya Suri (Dept. of Instrumentation and Applied Physics) Indian Institute of Science
 - Derived input-output relations for one-sided and two-sided cavity model for the readout of a transmon coupled to a resonator.
- **Design of optical detectors** May 2019 - July 2019
Supervisor: Dr. Ben Buchler (Dept. of Quantum Science) The Australian National University
 - Built a trans-impedance amplifier circuit for a photo-detector used for high speed detection of laser signals.
- **Study of structure of nano-confined water under electric field** May 2018 - April 2019
Supervisor: Prof. Prabal K Miati (Dept. of Physics) Indian Institute of Science
 - Performed modelling and molecular dynamics simulation of nano-confined water between two sheets of graphene, and studied its behaviour under electric field.
- **Computational identification of potential stop codon read-through candidates in mammalian cell** May 2018 - May 2019
Supervisor: Dr. Sandeep Eswarappa (Dept. of Biochemistry) Indian Institute of Science
 - Developed automated software suite to analyze large scale mass-spectrometric data for the identification of potential stop codon read-through candidates.

Transferable skills

- **Programming languages:** Proficient with Python and Julia.
- **Software tools:** QuTiP, JAX, Tensorflow, Git, $\LaTeX$, CI/CD.
- **Teamwork:** Working in a team of researchers and software engineer for the development of software package [ParaQeet](#).

Conferences and academic experience

- Presented a talk on **Gradient evaluation of analytic controls for large Hilbert space** at the American Physical Society (APS) Global physics summit in Anaheim, USA. March 2025
- Presented a talk on **Demolition measurement of transmon qubit** at the Deutsche Physikalische Gesellschaft in Berlin, Germany. March 2024
- Attended the 2022 Arnold Sommerfeld School on **Physics meets AI**, Munich, Germany. Sep 2022
- Teaching assistant for a course on **Quantum Field Theory** in Universität des Saarlandes. Oct 2021 - Feb 2022
- Presented poster at various conferences including the **SQA 2026, SQA 2025, AQuTe25, Quantum optimal control workshop 2024, WE-Heraeus seminar**, and various project meetings. 2022-2025
- Participated in Quantum Hackathons – **QHack 2023** and **QEC Hack ETH Zurich 2023**. 2023

Awards

- **Chennupati and Vidya Jagadish Visiting Scholarship and Fellowship Awards** May - July 2019
Awarded by The Australian National University (ANU) for a research internship at ANU.
This scholarship is awarded to up to 4 people each year.
- **Kishore Vaigyanik Protsahan Yojana (KVPY)** Aug 2016 - July 2021
Fellowship instituted by the Department of Science and Technology, Government of India.

References

- **Prof. Dr. Frank Wilhelm-Mauch**
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- **Dr. Alessandro Ciani**
Forschungszentrum Jülich GmbH, Institute for Quantum Computing Analytics (PGI-12), 52425, Jülich, Germany
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Publications

- [1] **A. Mishra**, E. Lupo, F. K. Wilhelm, and A. Ciani, *Analytical series expansion for efficient gradient evaluation in multi-qubit optimal control*, 2026. arXiv: [2607.26867](#) [[quant-ph](#)].
- [2] **A. Mishra**, A. Simm, M. Wald, A. Ciani, and N. Wittler, “ParaQeet: A quantum optimal control toolkit with simple parameter management”, *In preparation*, 2026.
- [3] **A. Mishra**, F. K. Wilhelm, and S. Machnes, *Demolition measurement protocol for transmon qubits*, 2025. arXiv: [2502.18322](#) [[quant-ph](#)].
- [4] S. Sahoo, **A. Mishra**, A. M. Diehl, and M. K. Jolly, “Dynamics of hepatocyte-cholangiocyte cell-fate decisions during liver development and regeneration”, *Iscience*, vol. 25, no. 9, 2022.

- [5] S. Sahoo, A. **Mishra**, H. Kaur, K. Hari, S. Muralidharan, S. Mandal, and M. K. Jolly, "A mechanistic model captures the emergence and implications of non-genetic heterogeneity and reversible drug resistance in er+ breast cancer cells", *NAR cancer*, vol. 3, no. 3, zcab027, 2021.
- [6] L. E. Manjunath, A. Singh, S. Sahoo, A. **Mishra**, J. Padmarajan, C. G. Basavaraju, and S. M. Eswarappa, "Stop codon readthrough of mammalian mtch2 leading to an unstable isoform regulates mitochondrial membrane potential", *Journal of Biological Chemistry*, jbc-RA120, 2020.